

Lecture 10: Universal and Perfect Hashing

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601.433/633 Introduction to Algorithms

Introduction

Another approach to dictionaries (insert, lookup, delete): hashing

- ▶ Can improve operations to $O(1)$, but with many caveats!

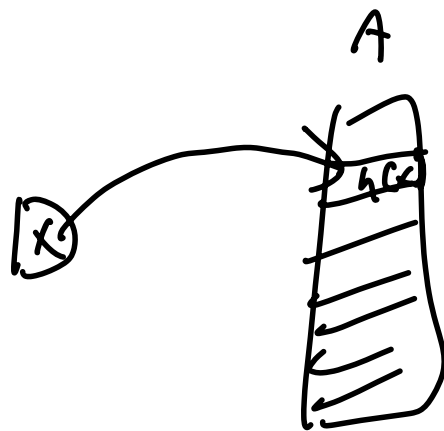
Should have seen some discussion of hashing in data structures. Also in CLRS.

- ▶ Separate chaining vs. open addressing

Today: discussion of caveats, more advanced versions of hashing (universal and perfect)

Hashing Basics

- ▶ Keys from universe U (think very large)
- ▶ Set $S \subseteq U$ of keys we actually care about (think relatively small). $|S| = N$.
- ▶ Hash table A (array) of size M .
- ▶ Hash function $h: U \rightarrow [M]$
 - ▶ $[M] = \{1, 2, \dots, M\}$
- ▶ Idea: store x in $A[h(x)]$

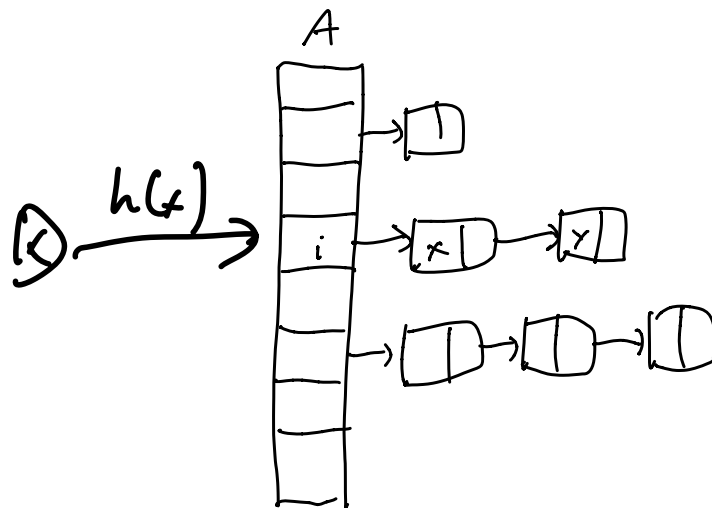


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One more component: *collision resolution*

- ▶ Today: *separate chaining*
- ▶ $A[i]$ is a linked list containing all x inserted where $h(x) = i$.



Dictionary Operations

Lookup(x): Walk down the list at $A[h(x)]$ until we find x (or walk to the end of the list)

Insert(x): Add x to the beginning of the list at $A[h(x)]$. $O(1)$

Delete(x): Walk down the list at $A[h(x)]$ until we find x . Remove it from the list.

$O(\text{length of } A[h(x)])$

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de minimis

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- ▶ h fast to compute.

Bad News

Theorem

For any hash function h , if $|U| \geq (N - 1)M + 1$, then there exists a set S of N elements that all hash to the same location.

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- ▶ Option 1: don't worry about it, hope world isn't adversarial.
- ▶ Option 2: Randomness! *Random function* $h: U \rightarrow [M]$
 - ▶ For each $x \in U$, choose $y \in [M]$ uniformly at random and set $h(x) = y$.
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 - ▶ Hopefully good behavior in expectation.
 - ▶ Problem: How can we store/remember/create h ?

Universal Hashing

Definition

A probability distribution $\mathbf{H}$ over hash functions $\{\mathbf{h} : \mathbf{U} \rightarrow [\mathbf{M}]\}$ is *universal* if

$$\Pr_{h \sim H}[h(x) = h(y)] \leq 1/M$$

for all $\mathbf{x}, \mathbf{y} \in \mathbf{U}$ with $\mathbf{x} \neq \mathbf{y}$.

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+1

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So Lookup($\mathbf{x}$) and Delete($\mathbf{x}$) have expected time $\mathbf{O}(1 + \mathbf{N}/\mathbf{M})$.

$\implies$ If $\mathbf{M} = \Omega(\mathbf{N})$, operations in $\mathbf{O}(1)$ time!

Linearity of Expectation

Amazing feature of expectations: linearity!

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For any two random variables $\mathbf{X}$ and $\mathbf{Y}$, and any constants α and β :

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$$\implies \mathbf{X} = \mathbf{X}_1 + \mathbf{X}_2. \text{ So } \mathbf{E}[\mathbf{X}] = \mathbf{E}[\mathbf{X}_1 + \mathbf{X}_2] = \mathbf{E}[\mathbf{X}_1] + \mathbf{E}[\mathbf{X}_2]$$

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$$\mathbf{E}[\mathbf{X}_1] = \mathbf{E}[\mathbf{X}_2] = \sum_{y=1}^6 \frac{1}{6}y = \frac{21}{6} = 3.5$$

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$$\implies \mathbf{E}[\mathbf{X}] = 3.5 + 3.5 = 7$$

Main Proof

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Proof.

$$\text{Let } C_{xy} = \begin{cases} 1 & \text{if } h(x) = h(y) \\ 0 & \text{otherwise} \end{cases}$$

$$\implies E[C_{xy}] = \Pr_{h \sim H}[h(x) = h(y)] \leq 1/M$$

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Number of collisions between $\mathbf{x}$ and $\mathbf{S}$ is exactly $\sum_{y \in \mathbf{S} \setminus \{x\}} C_{xy}$

$$\implies E \left[\sum_{y \in \mathbf{S} \setminus \{x\}} C_{xy} \right] = \sum_{y \in \mathbf{S} \setminus \{x\}} E[C_{xy}] \leq \sum_{y \in \mathbf{S} \setminus \{x\}} \frac{1}{M} = \mathbf{N}/\mathbf{M}$$

□

Main Corollary

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If $\mathbf{H}$ is universal, then for any sequence of $\mathbf{L}$ insert, lookup, and delete operations in which there are at most $\mathbf{O}(\mathbf{M})$ elements in the system at any time, the expected total cost of the whole sequence is only $\mathbf{O}(\mathbf{L})$ (assuming $\mathbf{h}$ takes constant time to compute).

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So universal distributions are great. Can we construct them?

Universal Hash Families

Definition

If $\mathcal{H}$ is universal and is a uniform distribution over a set of functions $\{h_1, h_2, \dots\}$, then that set is called a *universal hash family*.

Often use $\mathcal{H}$ to refer to both set of functions and uniform distribution over it.

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Notation:

- ▶ $\mathbf{U} = \{0, 1\}^u$ (so $|\mathbf{U}| = 2^u$)
- ▶ $\mathbf{M} = 2^b$, so an index to $\mathbf{A}$ is an element of $\{0, 1\}^b$

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Construction: $\mathbf{H} = \{0, 1\}^{b \times u}$, i.e., $\mathbf{H}$ is all $b \times u$ binary matrices

- ▶ Each $h \in \mathbf{H}$ is a (linear) function from $\mathbf{U}$ to $[\mathbf{M}]$:
 $h(x) = hx \in \{0, 1\}^b$ (all operations mod 2)

$$\begin{array}{c|c|c} h & x & h(x) \\ \hline \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix} & \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} & = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \end{array}$$

Universality

Theorem

***H** is a universal hash family: $\Pr_{h \sim H}[h(x) = h(y)] \leq 1/M$ for all $x \neq y \in \{0, 1\}^u$.*

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Since $x \neq y$, there is i s.t. $x_i \neq y_i$. WLOG, $x_i = 0$ and $y_i = 1$.

The diagram shows two horizontal vectors, x and y , enclosed in red hand-drawn ovals. Vector x has an ellipsis on the left, a vertical line, a box containing '0', another vertical line, and an ellipsis on the right. Vector y has an ellipsis on the left, a vertical line, a box containing '1', another vertical line, and an ellipsis on the right. The labels x and y are to the right of their respective ovals.

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Draw all entries of h except for h^i . Let $h' = h$ with h^i all 0's

- ▶ $h(x) = h'(x)$ already fixed.

$$h = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 \end{bmatrix}$$

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Handwritten red equations illustrating the proof logic:

$$\begin{matrix} h^i & & h^i \\ \left[\begin{smallmatrix} \vdots \\ 0 \end{smallmatrix} \right] & & \left[\begin{smallmatrix} \vdots \\ c \\ 0 \end{smallmatrix} \right] \end{matrix} \Rightarrow \left[\begin{smallmatrix} c \\ \vdots \\ 0 \end{smallmatrix} \right]$$

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- ▶ $h(x) = h'(x)$ already fixed.
- ▶ If $h(y) = h(x)$, then h^i must equal $h(x) - h'(y)$
- ▶ Happens with probability exactly $1/2^b = 1/M$



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Can we do better with hashing? Yes, through universal hashing!

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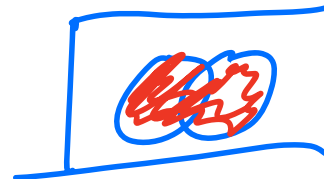
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$\Pr_{h \sim H}[h(x) = h(y)] \leq 1/M = 1/N^2$ by universality.

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$$\begin{aligned} \Pr_{h \sim H}[\exists \text{ collision in } S] &\leq \sum_{\{x, y\} \in \binom{S}{2}} \Pr_{h \sim H}[h(x) = h(y)] \leq \sum_{\substack{x, y \in S \\ x \neq y}} \frac{1}{N^2} \\ &= \binom{N}{2} \frac{1}{N^2} = \frac{N(N-1)}{2} \frac{1}{N^2} \leq \frac{1}{2} \end{aligned}$$



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So keep sampling $h \sim H$ until get one with no collisions!

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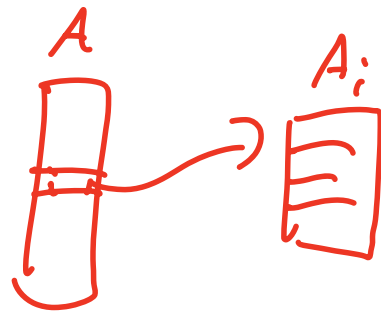
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For each $i \in [M]$, let $S_i = \{x \in S : h(x) = i\}$ and let $n_i = |S_i|$

- ▶ Use another hash table for S_i !
- ▶ Use Method 1: $O(n_i^2)$ -size perfect hashing of S_i .
 - ▶ Let $h_i : U \rightarrow [n_i^2]$ be hash function for S_i , and A_i be table (pointer from $A[i]$)

Method 2

$M = N^2$ is pretty big!

- ▶ Only storing N things, and know them ahead of time
- ▶ Want space $O(N)$
- ▶ Open question for a long time!

Starting approach: set $M = N$, use a universal hash family H . Draw $h \sim H$.

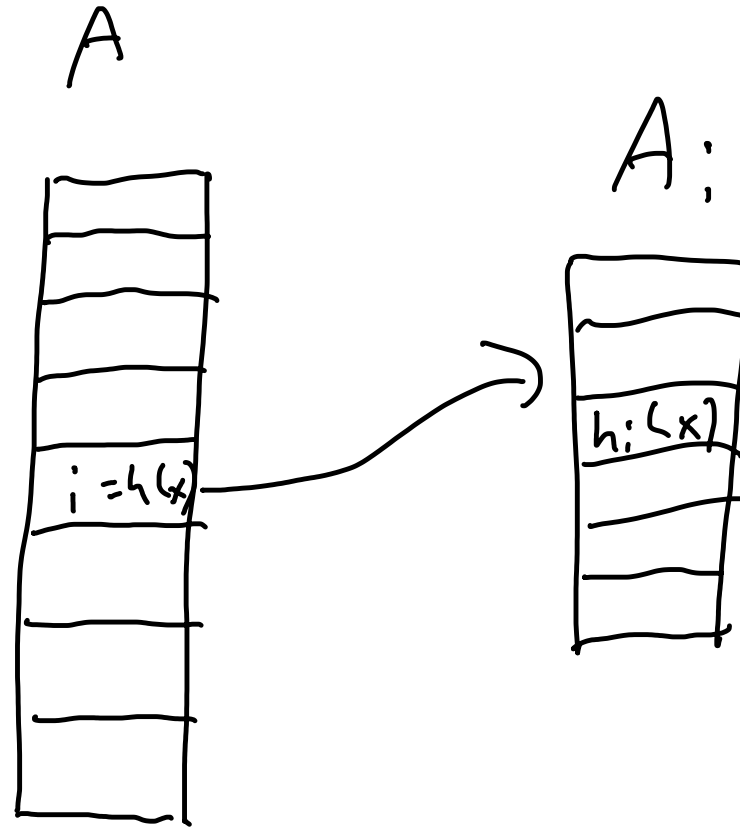
- ▶ Will have collisions. Need to do something other than chaining.

For each $i \in [M]$, let $S_i = \{x \in S : h(x) = i\}$ and let $n_i = |S_i|$

- ▶ Use another hash table for S_i !
- ▶ Use Method 1: $O(n_i^2)$ -size perfect hashing of S_i .
 - ▶ Let $h_i : U \rightarrow [n_i^2]$ be hash function for S_i , and A_i be table (pointer from $A[i]$)

Lookup(x): Look in $A_{h(x)}[h_{h(x)}(x)]$

Picture



Analysis

Lookup time: by analysis of Method 1, no collisions in second level.

⇒ Lookup time **$O(1)$**

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Theorem

Let H be universal onto a table of size N . Then

$$\Pr_{h \sim H} \left[\sum_{i=1}^N n_i^2 > 4N \right] < 1/2.$$

So like with method 1: keep drawing $h \sim H$ until $\sum_{i=1}^N n_i^2 \leq 4N$

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Prove that $E \left[\sum_{i=1}^N n_i^2 \right] \leq 2N$.

- ▶ Implies theorem by Markov's inequality
 - ▶ $\Pr[X > 2E[X]] \leq 1/2$ for nonnegative random variables X .

Proof

Observation: $\sum_{i=1}^N n_i^2$ is exactly number of *ordered* pairs that collide, including self-collisions

- ▶ Example: If $S_i = \{\mathbf{a}, \mathbf{b}, \mathbf{c}\}$ then $n_i^2 = \mathbf{9}$. Ordered colliding pairs:
 $(\mathbf{a}, \mathbf{a}), (\mathbf{a}, \mathbf{b}), (\mathbf{a}, \mathbf{c}), (\mathbf{b}, \mathbf{a}), (\mathbf{b}, \mathbf{b}), (\mathbf{b}, \mathbf{c}), (\mathbf{c}, \mathbf{a}), (\mathbf{c}, \mathbf{b}), (\mathbf{c}, \mathbf{c})$

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$$\text{Let } C_{xy} = \begin{cases} 1 & \text{if } h(x) = h(y) \\ 0 & \text{otherwise} \end{cases}$$

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$$\text{Let } C_{xy} = \begin{cases} 1 & \text{if } h(x) = h(y) \\ 0 & \text{otherwise} \end{cases}$$

$$\begin{aligned} E \left[\sum_{i=1}^N n_i^2 \right] &= E \left[\sum_{x \in S} \sum_{y \in S} C_{xy} \right] \\ &= N + \sum_{x \in S} \sum_{y \in S: y \neq x} E[C_{xy}] && \text{(linearity of expectations)} \\ &\leq N + \frac{N(N-1)}{M} && \text{(definition of universal)} \\ &< 2N && \text{(since } M = N) \end{aligned}$$